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## BURST MODE OPERATION OF INTRAVASCULAR LITHOTRIPSY (IVL)

### Abstract

Disclosed herein are systems and methods for applying voltage(s) across electrodes of a shock wave catheter system. The electrodes in the shock wave catheter may be activated by using at least one packet comprising a plurality of sub-pulses. The plurality of sub-pulses within a packet may be delivered in rapid succession (e.g., with a frequency of 100 Hz-10 kHz, pulse width duration of 1 μs or shorter, and/or peak power of 250 kW or higher). The packets may be delivered in bursts with, e.g., a frequency of 1-4 Hz). The time between adjacent sub-pulses may be less than the time between adjacent packets. The sub-pulses may be delivered with a higher peak power (lower energy) compared to non-burst pulses. The voltages of the sub-pulses may be applied using a single- or multi-stage generator circuit. The shock wave catheter may be operated in burst mode operation and/or non-burst mode operation.

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## Background/Summary

FIELD OF THE DISCLOSURE

[0001] The present disclosure relates generally to the field of medical devices and methods, and more specifically to shock wave catheter devices for treating calcified lesions in body lumens, such as calcified lesions and occlusions in vasculature and kidney stones in the urinary system.

BACKGROUND

[0002] A wide variety of catheters have been developed for treating calcified lesions, such as calcified lesions in vasculature associated with arterial disease. For example, treatment systems for percutaneous coronary angioplasty or peripheral angioplasty use angioplasty balloons to dilate a calcified lesion and restore normal blood flow in a vessel. In these types of procedures, a catheter carrying a balloon is advanced into the vasculature along a guide wire until the balloon is aligned with calcified plaques. The balloon is then pressurized (normally to greater than 10 atm), causing the balloon to expand in a vessel to push calcified plaques back into the vessel wall and dilate occluded regions of vasculature.

[0003] More recently, the technique and treatment of intravascular lithotripsy (IVL) has been developed, which is an interventional procedure to modify calcified plaque in diseased arteries. The mechanism of plaque modification is through use of a catheter having one or more acoustic shock wave generating sources located within a liquid that can generate acoustic shock waves that modify the calcified plaque. IVL devices vary in design with respect to the energy source used to generate the acoustic shock waves, with two exemplary energy sources being electrohydraulic generation and laser generation.

[0004] For electrohydraulic generation of acoustic shock waves, a conductive solution (e.g., saline) may be contained within an enclosure that surrounds electrodes or can be flushed through a tube that surrounds the electrodes. The calcified plaque modification is achieved by creating acoustic shock waves within the catheter by an electrical discharge across the electrodes. The energy from this electrical discharge enters the surrounding fluid faster than the speed of sound, generating an acoustic shock wave. In addition, the energy creates one or more rapidly expanding and collapsing vapor bubbles that generate secondary shock waves. The shock waves propagate radially outward and modify calcified plaque within the blood vessels. For laser generation of acoustic shock waves, a laser pulse is transmitted into and absorbed by a fluid within the catheter. This absorption process rapidly heats and vaporizes the fluid, thereby generating the rapidly expanding and collapsing vapor bubble, as well as the acoustic shock waves that propagate outward and modify the calcified plaque. The acoustic shock wave intensity is higher if a fluid is chosen that exhibits strong absorption at the laser wavelength that is employed. These examples of IVL devices are not intended to be a comprehensive list of potential energy sources to create IVL shock waves.

[0005] The IVL process may be considered different from standard atherectomy procedures in that it cracks calcium but does not liberate the cracked calcium from the tissue. Hence, generally speaking, IVL should not require aspiration nor embolic protection. Further, due to the compliance of a normal blood vessel and non-calcified plaque, the shock waves produced by IVL do not modify the normal vessel tissue or non-calcified plaque. Moreover, IVL does not carry the same degree of risk of perforation, dissection, or other damage to vasculature as atherectomy procedures or angioplasty procedures using cutting or scoring balloons.

[0006] More specifically, catheters to deliver IVL therapy have been developed that include pairs of electrodes for electrohydraulically generating shock waves inside an angioplasty balloon. Shock wave devices can be particularly effective for treating calcified plaque lesions because the acoustic pressure from the shock waves can crack and disrupt lesions near the angioplasty balloon without harming the surrounding tissue. In these devices, the catheter is advanced over a guidewire through a patient's vasculature until it is positioned proximal to and/or aligned with a calcified plaque lesion in a body lumen. The balloon is then inflated with conductive fluid (using a relatively low pressure of 2-4 atm) so that the balloon expands to contact the lesion, but is not an inflation pressure that substantively displaces the lesion. Voltage pulses can then be applied across the electrodes of the electrode pairs to produce acoustic shock waves that propagate through the walls of the angioplasty balloon and into the lesions. Once the lesions have been cracked by the acoustic shock waves, the balloon can be expanded further to increase the cross-sectional area of the lumen and improve blood flow through the lumen. Alternative devices to deliver IVL therapy can be within a closed volume other than an angioplasty balloon, such as a cap, balloons of various compliances, or other enclosures.

[0007] The voltage pulses may be non-burst pulses applied across the electrodes of the electrode pairs and may be effective at producing the shock waves to treat, e.g., calcium lesions. A conductive solution (e.g., saline) may be contained within an enclosure that surrounds the electrodes or can be flushed through a tube that surrounds the electrodes. Acoustic shock waves may be created within the catheter by an electrical discharge across the electrodes. These shock waves propagate radially outward and modify calcified plaque. In some embodiments, laser generation of acoustic shock waves is used, where a laser pulse is transmitted into and absorbed by a fluid within the catheter. This absorption process rapidly heats and vaporizes the fluid, thereby generating the rapidly expanding vapor bubble, as well as the acoustic shock waves that propagate outward and modify the calcified plaque. More effective techniques for delivering shock wave therapy may be desirable.

SUMMARY

[0008] Described herein are systems and methods for applying one or more voltages across electrodes of a shock wave catheter system. The electrodes in the shock wave catheter may be activated by using at least one packet, which comprises a plurality of sub-pulses. The plurality of sub-pulses within a packet may be delivered in rapid succession (e.g., with a frequency of 100 Hz-10 kHz), and the packets may be delivered in bursts. In some embodiments, the time between adjacent sub-pulses may be less than the time between adjacent packets. The packets may have a frequency of 1-4 Hz, for example. Embodiments of the disclosures may comprise applying sub-pulses having a higher peak power (but lower energy) to the electrodes of the shock wave catheter system compared to non-burst pulses.

[0009] A method for generating one or more shock waves in a shock wave catheter system is disclosed. The method comprises: applying a first voltage to one or more electrodes of the shock wave catheter system to generate one or more bubbles in a fluid surrounding the one or more electrodes; and applying at least one packet of one or more second voltages to the one or more electrodes to generate one or more electrical arcs at the one or more electrodes, wherein: each of the at least one packet comprises a plurality of sub-pulses, and at least one of the plurality of sub-pulses has a duration of 1 μs or shorter, or a frequency of the plurality of sub-pulses is between 100 Hz to 10 KHz; wherein the one or more second voltages are different from the first voltage. Additionally or alternatively, in some embodiments, properties of the plurality of sub-pulses are based on properties of calcium and/or tissue treated by the shock wave catheter system, wherein the properties of the calcium and/or tissue comprise a hardness, thickness, acoustic properties, or a combination thereof. Additionally or alternatively, in some embodiments, the properties of the plurality of sub-pulses include a number of the plurality of sub-pulses within the at least one packet, a duration of the at least one sub-pulse, a peak power of the at least one sub-pulse, a frequency of the plurality of sub-pulses, an electrical pulse amplitude, a sonic output, or a combination thereof. Additionally or alternatively, in some embodiments, a number of the plurality of sub-pulses is greater than or equal to 10. Additionally or alternatively, in some embodiments, the at least one packet has a duration of 20 μs or longer, a duty cycle of 50%, or both. Additionally or alternatively, in some embodiments, a plurality of packets has a frequency between 1 to 4 Hz, wherein the plurality of packets includes the at least one packet. Additionally or alternatively, in some embodiments, a frequency of the plurality of sub-pulses is 100 times more than a frequency of a plurality of packets, wherein the plurality of packets includes the at least one packet. Additionally or alternatively, in some embodiments, the method further comprises: applying one or more third voltages during one or more non-active sub-pulse periods between the plurality of sub-pulses. Additionally or alternatively, in some embodiments, the one or more non-active sub-pulse periods have a power level that is 50% or less than a power level of the plurality of sub-pulses. Additionally or alternatively, in some embodiments, a peak power of the at least one sub-pulse is 250 kW or higher.

[0010] A shock wave catheter system is disclosed. The shock wave catheter system comprises: one or more electrodes; a fluid surrounding the one or more electrodes; a first voltage source configured to apply a first voltage to the one or more electrodes to generate one or more bubbles in the fluid; and a second voltage source configured to apply at least one packet of one or more second voltages to the one or more electrodes to generate one or more electrical arcs at the one or more electrodes, wherein the at least one packet comprises a plurality of sub-pulses, wherein at least one of the plurality of sub-pulses has a duration of 1 μs or shorter, or a frequency of the plurality of sub-pulses is between 100 Hz to 10 kHz. Additionally or alternatively, in some embodiments, the shock wave catheter system further comprises: a controller configured to control properties of the plurality of sub-pulses based on properties of calcium and/or tissues treated by the shock wave catheter system, wherein the properties of the calcium and/or tissue comprise a hardness, thickness, acoustic properties, or a combination thereof. Additionally or alternatively, in some embodiments, the properties of the plurality of sub-pulses include a number of the plurality of sub-pulses within the at least one packet, a duration of the at least one sub-pulse, a peak power of the at least one sub-pulse, an electrical pulse amplitude, a sonic output, or a combination thereof. Additionally or alternatively, in some embodiments, a number of the plurality of sub-pulses is greater than or equal to 10. Additionally or alternatively, in some embodiments, the at least one packet has a duration of 20 μs or longer, a duty cycle of 50%, or both. Additionally or alternatively, in some embodiments, a plurality of packets has a frequency between 1 to 4 Hz, wherein the plurality of packets includes the at least one packet. Additionally or alternatively, in some embodiments, the first voltage source is a low power voltage source, and the second voltage source is a high-power voltage source. Additionally or alternatively, in some embodiments, the one or more electrodes comprise an electrode pair separated by an electrode gap and the one or more second voltages are applied across the electrode gap.

[0011] A method for generating one or more shock waves in a shock wave catheter system is disclosed. The method comprises: applying a first voltage to one or more electrodes of the shock wave catheter system to generate one or more bubbles in a fluid surrounding the one or more electrodes; and applying a plurality of sub-pulses of one or more second voltages to the one or more electrodes to generate one or more electrical arcs at the one or more electrodes, wherein the plurality of sub-pulses has a frequency greater than 10 Hz.

[0012] A shock wave catheter system is disclosed. A shock wave catheter system comprises: one or more electrodes; a fluid surrounding the one or more electrodes; a first voltage source configured to apply a first voltage to the one or more electrodes to generate one or more bubbles in the fluid; and a second voltage source configured to apply a plurality of sub-pulses of one or more second voltages to the one or more electrodes to generate one or more electrical arcs at the one or more electrodes, wherein the plurality of sub-pulses has a frequency greater than 10 Hz.

[0013] A method for generating one or more shock waves in a shock wave catheter system is disclosed. The method comprises: applying a first voltage to one or more electrodes of the shock

wave catheter system to generate one or more bubbles in a fluid surrounding the one or more electrodes; and applying a plurality of packets of one or more second voltages to the one or more electrodes to generate one or more electrical arcs at the one or more electrodes, wherein the plurality of packets has a frequency greater than 1 Hz, wherein applying the plurality of packets of the one or more second voltages comprises: for each of the plurality of packets, applying a plurality of sub-pulses of the one or more second voltages to the one or more electrodes.
[0014] A shock wave catheter system is disclosed. The shock wave catheter system comprises: one or more electrodes; a fluid surrounding the one or more electrodes; a first voltage source configured to apply a first voltage to the one or more electrodes to generate one or more bubbles in the fluid; and a second voltage source configured to apply a plurality of packets of one or more second voltages to the one or more electrodes to generate one or more electrical arcs at the one or more electrodes, wherein the plurality of packets has a frequency greater than 1 Hz, wherein the second voltage source applying the plurality of packets of the one or more second voltages comprises: for each of the plurality of packets, applying a plurality of sub-pulses of the one or more second voltages to the one or more electrodes.
[0015] A method for operating a shock wave catheter system is disclosed. The method comprises: applying a first voltage to one or more electrodes of the shock wave catheter system to generate one or more bubbles in a fluid surrounding the one or more electrodes; charging a reservoir capacitor using a second voltage source; charging a sub-pulse capacitor using the reservoir capacitor; and for a plurality of sub-pulses included in a packet, delivering energy stored in the sub-pulse capacitor to the one or more electrodes to generate one or more electrical arcs at the one or more electrodes.
[0016] A circuit is disclosed. The circuit comprises: a first voltage source configured to apply a first voltage to one or more electrodes of a shock wave catheter system to generate one or more bubbles in a fluid surrounding the one or more electrodes; a second voltage source configured to apply a second voltage; a reservoir capacitor coupled to the second voltage source and configured to store a charge from the second voltage source; and a sub-pulse capacitor coupled to the reservoir capacitor, wherein the sub-pulse capacitor is configured to store a charge from the reservoir capacitor and transfer stored energy to the one or more electrodes to generate one or more electrical arcs at the one or more electrodes.
[0017] A shock wave catheter system is disclosed. The shock wave catheter system comprises: one or more electrodes; a fluid surrounding the one or more electrodes; a first voltage source configured to apply a first voltage to the one or more electrodes to generate one or more bubbles in the fluid; a second voltage source configured to apply a second voltage; a reservoir capacitor coupled to the second voltage source and configured to store a charge from the second voltage source; and a sub-pulse capacitor coupled to the reservoir capacitor, wherein the sub-pulse capacitor is configured to store a charge from the reservoir capacitor and transfer stored energy to the one or more electrodes to generate one or more electrical arcs at the one or more electrodes.
[0018] A method for operating a shock wave catheter system is disclosed. The method comprises: applying a first voltage to one or more electrodes of the shock wave catheter system to generate one or more bubbles in a fluid surrounding the one or more electrodes; charging a capacitor using a second voltage source; and for a plurality of sub-pulses included in a packet, delivering energy stored in the capacitor to the one or more electrodes to generate one or more electrical arcs at the one or more electrodes, wherein at least one of the plurality of sub-pulses has a duration of 1 μs or shorter, or a frequency of the plurality of sub-pulses is between 100 Hz to 10 kHz.
[0019] A circuit is disclosed. The circuit comprises: a first voltage source configured to apply a first voltage to one or more electrodes of a shock wave catheter system to generate one or more bubbles in a fluid surrounding the one or more electrodes; a second voltage source configured to apply a second voltage comprising a plurality of sub-pulses; and a capacitor coupled to the second voltage source, wherein the capacitor is configured to store a charge from the second voltage source and

transfer stored energy to the one or more electrodes to generate one or more electrical arcs at the one or more electrodes, wherein at least one of the plurality of sub-pulses has a duration of 1 μs or shorter, or a frequency of the plurality of sub-pulses is between 100 Hz to 10 kHz.

[0020] A shock wave catheter system is disclosed. The shock wave catheter system comprises: one or more electrodes; a fluid surrounding the one or more electrodes; a first voltage source configured to apply a first voltage to the one or more electrodes to generate one or more bubbles in the fluid; a second voltage source configured to apply a second voltage comprising a plurality of sub-pulses; a capacitor coupled to the second voltage source, wherein the capacitor is configured to store a charge from the second voltage source and transfer stored energy to the one or more electrodes to generate one or more electrical arcs at the one or more electrodes, wherein at least one of the plurality of sub-pulses has a duration of 1 μs or shorter, or a frequency of the plurality of sub-pulses is between 100 Hz to 10 KHz.

[0021] A method of treating a lesion in a body lumen is disclosed. The method comprises: advancing a shock wave catheter to the lesion through the body lumen; and applying a packet of voltage pulses, the packet comprising a plurality of sub-pulses delivered at a frequency of at least 10 Hz, wherein each of the plurality of sub-pulses generates a shock wave. Additionally or alternatively, in some embodiments, properties of the plurality of sub-pulses are based on properties of calcium and/or tissue treated by the shock wave catheter, wherein the properties of the calcium and/or tissue comprise a hardness, thickness, acoustic properties, or a combination thereof. Additionally or alternatively, in some embodiments, the properties of the plurality of sub-pulses include a number of the plurality of sub-pulses within the at least one packet, a duration of the at least one sub-pulse, a peak power of the at least one sub-pulse, a frequency of the plurality of sub-pulses, an electrical pulse amplitude, a sonic output, or a combination thereof. Additionally or alternatively, in some embodiments, a number of the plurality of sub-pulses is greater than or equal to 10. Additionally or alternatively, in some embodiments, the packet has a duration of 20 μs or longer, a duty cycle of 50%, or both. Additionally or alternatively, in some embodiments, a plurality of packets has a frequency between 1 to 4 Hz, wherein the plurality of packets includes the packet. Additionally or alternatively, in some embodiments, a frequency of the plurality of sub-pulses is 100 times more than a frequency of a plurality of packets, wherein the plurality of packets includes the packet. Additionally or alternatively, in some embodiments, the method further comprises: applying one or more second voltages during one or more non-active sub-pulse periods between the plurality of sub-pulses. Additionally or alternatively, in some embodiments, the one or more non-active sub-pulse periods have a power level that is 50% or less than a power level of the plurality of sub-pulses. Additionally or alternatively, in some embodiments, a peak power of at least one of the plurality of sub-pulses is 250 kW or higher.

[0022] A shock wave catheter system is disclosed. The shock wave catheter system comprises: one or more electrodes; a fluid surrounding the one or more electrodes; and one or more voltage sources configured to apply one or more voltages to the one or more electrodes in accordance with a mode of operation, wherein the mode of operation comprises a burst mode operation and a non-burst mode operation, wherein the burst mode operation comprises applying at least one packet of the one or more voltages to the one or more electrodes, each of the at least one packet comprises a plurality of sub-pulses, and at least one of the plurality of sub-pulses has a duration of 1 μs or shorter, or a frequency of the plurality of sub-pulses is between 100 Hz to 10 KHz.

[0023] A method for operating a shock wave catheter system is disclosed. A method for operating a shock wave catheter system, comprising: operating the shock wave catheter system in a burst mode operation, wherein the burst mode operation comprises applying at least one packet of one or more voltages to one or more electrodes of the shock wave catheter system, wherein: each of the at least one packet comprises a plurality of sub-pulses, and at least one of the plurality of sub-pulses has a duration of 1 μs or shorter, or a frequency of the plurality of sub-pulses is between 100 Hz to 10 kHz; and operating the shock wave catheter system in a non-burst mode operation.

## Description

DESCRIPTION OF THE FIGURES
[0024] The invention will now be described, by way of example only, with reference to the accompanying drawings, in which:
[0025] FIG. **1** illustrates a side view of an exemplary angioplasty balloon catheter, according to some embodiments.
[0026] FIG. **2** illustrates a side view of example electrodes electrically coupled to a voltage source, according to some embodiments.
[0027] FIG. **3** illustrates example inner and outer electrodes, according to some embodiments.
[0028] FIG. **4** illustrates an exemplary simplified equivalent circuit diagram, according to some embodiments.
[0029] FIG. **5** illustrates an exemplary graph of a pulse applied to the electrodes and the resulting current flow through the electrodes, according to some embodiments.
[0030] FIG. **6**A illustrates an expanded view of an example packet comprising a plurality of sub-pulses delivered in rapid succession, according to some embodiments.
[0031] FIG. **6**B illustrates exemplary graphs of a plurality of sub-pulses (top) and non-burst pulses (bottom), according to some embodiments.
[0032] FIG. **7** illustrates an exemplary graph comparing non-burst (non-packet) pulses (top) and a plurality of sub-pulses (bottom), according to some embodiments.
[0033] FIG. **8**A illustrates an example comparison of power and energy (area under the power vs. time curve) for a packet comprising sub-pulses and a non-burst pulse, according to some embodiments.
[0034] FIG. **8**B illustrates an example comparison of power and energy for a packet comprising sub-pulses and a non-burst pulse, according to some embodiments.
[0035] FIG. **9** illustrates a schematic diagram of an example dual stage generator circuit, according to some embodiments.
[0036] FIG. **10** illustrates an exemplary flow chart for burst mode operation of an IVL catheter, according to some embodiments.
[0037] FIG. **11** illustrates a graph of an exemplary burst mode operation of a multi-stage generator circuit for a packet, according to some embodiments.
[0038] FIG. **12** illustrates a graph of exemplary charge and sub-pulse modes, according to some embodiments.
[0039] FIG. **13** illustrates an exemplary sub-pulse, according to some embodiments.
[0040] FIG. **14** illustrates a simplified schematic diagram of an example single-stage generator circuit, according to some embodiments.
[0041] FIG. **15** illustrates a graph of an exemplary burst mode operation of a single-stage generator circuit for a packet, according to some embodiments.
[0042] FIG. **16** illustrates an exemplary sub-pulse, according to some embodiments.
[0043] FIG. **17** illustrates an exemplary of a computing system, according to some embodiments.
DETAILED DESCRIPTION
[0044] The following description is presented to enable a person of ordinary skill in the art to make and use the various embodiments and aspects thereof disclosed herein. Descriptions of specific devices, assemblies, techniques, and applications are provided only as examples. Various modifications to the examples described herein will be readily apparent to those of ordinary skill in the art, and the general principles described herein may be applied to other examples and applications without departing from the spirit and scope of the various embodiments and aspects thereof. Thus, the various embodiments and aspects thereof are not intended to be limited to the examples described herein and shown but are to be accorded the scope consistent with the claims.

[0045] Efforts have been made to improve the design of electrode assemblies included in shock wave and directed cavitation catheters. For instance, low-profile electrode assemblies have been developed that reduce the crossing profile of a catheter and allow the catheter to more easily navigate calcified vessels to deliver shock waves in more severely occluded regions of vasculature. Examples of low-profile electrode designs can be found in U.S. Pat. Nos. 8,888,788, 9,433,428, and 10,709,462, and in U.S. Publication No. 2021/0085383, all of which are incorporated herein by reference. Other catheter designs have improved the delivery of shock waves, for instance, by specific electrode construction and configuration thereby directing shock waves in a forward direction to break up tighter and harder-to-cross occlusions in vasculature. Examples of forward-firing catheter designs can be found in U.S. Pat. Nos. 10,966,737, 11,478,261, and 11,596,423 and U.S. Publication Nos. 2023/0107690 and 2023/0165598, all of which are incorporated herein by reference. Efforts to improve the longevity of electrode assemblies have included switching or alternating the polarity of the voltage source. Examples of polarity switching in a shock wave device can be found in U.S. Pat. No. 10,226,265, which is incorporated herein by reference.
[0046] In the following description of the various embodiments, reference is made to the accompanying drawings, in which are shown, by way of illustration, specific embodiments that can be practiced. It is to be understood that other embodiments and examples can be practiced, and changes can be made without departing from the scope of the disclosure.
[0047] In addition, it is also to be understood that the singular forms “a,” “an,” and “the” used in the following description are intended to include the plural forms as well, unless the context clearly indicates otherwise. It is also to be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It is further to be understood that the terms “includes, “including,” “comprises,” and/or “comprising,” when used herein, specify the presence of stated features, integers, steps, operations, elements, components, and/or units but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, units, and/or groups thereof. As provided herein, it should be appreciated that any disclosure of a numerical range describing dimensions or measurements such as thicknesses, length, weight, time, frequency, temperature, voltage, current, angle, etc. is inclusive of any numerical increment or gradient within the ranges set forth relative to the given dimension or measurement.
[0048] FIG. **1** illustrates a side view of an exemplary angioplasty balloon catheter, according to some embodiments. Catheter **20** includes an elongated carrier such as a hollow sheath **21**, a dilating balloon **26** formed about a sheath **21** in sealed relation thereto, and a guide wire member **28** to which the balloon **26** is scaled at a seal **23**. The guide wire member **28** may have a longitudinal lumen **29** through which a guide wire (not shown) may be received for directing the catheter **20** to a desired location within a vein or artery, for example.
[0049] The sheath **21** forms, with the guide wire member **28**, a channel **27** through which fluid may be admitted into the balloon **26** to inflate the balloon. The balloon **26** may be filled with water, saline, a mixed saline solution, etc., allowing the balloon to be gently placed along the walls of the artery or vein, for example, in direct proximity with the calcified lesion. The fluid may also contain an x-ray contrast to permit fluoroscopic viewing of the catheter during use. The channel **27** is in fluid communication with an electrode pair **25** and provides the conductive fluid necessary for current to flow across an electrode gap of the electrode pair **25** and subsequent shock wave generation. The electrode pair **25** may include electrodes **22** and **24** within the fluid filled balloon **26**.
[0050] Although FIG. **1** illustrates and the corresponding descriptions are specific to an intravascular lithotripsy (IVL) angioplasty balloon catheter, embodiments of the disclosure are applicable to other types of IVL, where a balloon may or may not be used. Additionally or alternatively, embodiments of the disclosure may include a plurality of electrode pairs **25**.
[0051] In some embodiments, an IVL catheter is a so-called “rapid exchange-type” (“Rx”) catheter

provided with an opening portion through which a guide wire is guided (e.g., through a middle portion of a central tube in a longitudinal direction). In other embodiments, an IVL catheter may be an "over-the-wire-type" ("OTW") catheter in which a guide wire lumen is formed throughout the overall length of the catheter, and a guide wire is guided through the proximal end of a hub.

[0052] As shown in FIG. **2**, the electrodes **22** and **24** may be electrically coupled to a source **40** (e.g., a voltage source). The electrodes **22** and **24** may be electrically coupled to the source **40** through a connector, for example. In some embodiments, the source **40** may be a high voltage power supply (HVPS). The source **40** may apply one or more voltages to the electrode **22**, the electrode **24**, or both to generate one or more electrical arcs at the electrodes.

[0053] As may be seen in FIG. **3**, the electrodes **22** and **24** are coaxially disposed with electrode **22** being an inner electrode and electrode **24** being an outer electrode. Referring back to FIG. **2**, the inner electrode **22** may be coupled to a positive terminal **44** of the source **40**, and the outer electrode **24** may be coupled to a negative terminal **46** of the source **40**. In some aspects, the system may employ polarity switching, where the inner electrode **22** may be coupled to the negative terminal **46** of the source, and the outer electrode **24** may be coupled to the positive terminal **44** of the source **40**. The electrodes **22** and **24** may be formed of metal, such as stainless steel, or another conductive material. The electrodes **22** and **24** may be spatially separated by a certain distance to allow a reproducible arc to form for a given applied voltage and current.

[0054] The electrical arcs between electrodes **22** and **24** in the fluid are used to generate shock waves in the fluid (within channel **27**). A pulse of high voltage applied to the electrodes **22** and **24** may form an arc across the electrodes **22** and **24**. Once the catheter **20** is positioned at a lesion site with the guide wire (not shown), the physician or operator can start applying pulses to the electrodes **22** and **24** (e.g., by pressing a button that controls the catheter system) to form the shock waves that crack the calcified plaque. Such shock waves may propagate through the fluid, through the balloon **26**, through the blood and vessel wall to the calcified lesion where the energy may break the hardened plaque without the application of excessive pressure by the balloon on the walls of the artery.

[0055] The device **200** may include multiple electrode pairs along the length of the balloon **26**. A shock wave device having multiple shock wave electrode pairs at different locations (circumferentially and/or longitudinally) may help to provide consistent or uniform force to a region of tissue. The electrode pairs may be electrically coupled in series or in parallel.

[0056] As used herein, the term "electrode" refers to an electrically conducting element (typically made of metal) that receives electrical current and subsequently releases the electrical current to another electrically conducting element. In the context of the present disclosure, electrodes are often positioned relative to each other, such as in an arrangement of an inner electrode and an outer electrode. Accordingly, as used herein, the term "electrode pair" refers to two electrodes that are positioned adjacent to and spaced apart from each other such that application of a sufficiently high voltage to the electrode pair will cause an electrical current to transmit across the gap (also referred to as a "spark gap") between the two electrodes (e.g., from an inner electrode to an outer electrode, or vice versa, optionally with the electricity passing through a conductive fluid or gas therebetween). In some contexts, one or more electrode pairs may also be referred to as an electrode assembly. In the context of the present disclosure, the term "emitter" broadly refers to the region of an electrode assembly where the current transmits across the electrode pair, generating a shock wave. The terms "emitter sheath" and "emitter band" refer to a continuous or discontinuous band of conductive material that may form one or more electrodes of one or more electrode pairs, thereby forming a location of one or more emitters.

[0057] FIG. **4** illustrates an exemplary simplified equivalent circuit diagram, according to some embodiments. The circuit may include a capacitor **50** that stores a voltage Vc. A switch **60** may be closed, allowing a voltage drop across the electrodes **22** and **24**.

[0058] FIG. **5** illustrates an exemplary graph of a pulse applied to the electrodes **22** and **24** and the resulting current flow through the electrodes **22** and **24**, according to some embodiments. The switch **60** may be first closed at time T**1**. At time T**1**, the voltage is low, and closing the switch **60** causes the voltage across the electrodes **22** and **24** to quickly rise to a voltage level **70**. During this time, the current **72** through the electrodes **22** and **24** is relatively low. After a dwell time Td, when the voltage across the electrodes **22** and **24** reaches the breakdown voltage of the liquid between the electrodes **22** and **24**, an electrical arc occurs across the electrodes **22** and **24**. After the dwell time Td, at time T**2**, the electrical arc causes a high current **72** to begin to flow through the electrodes **22** and **24** and a plasma to form between the electrodes **22** and **24**. The plasma rapidly heats the liquid and creates a shock wave because the energy is transferred to the liquid faster than the speed of sound. In addition, the energy transfer to the liquid rapidly heats and vaporizes fluid thereby forming a vapor bubble. The bubble expands and then is cooled by the denser surrounding liquid and ultimately collapses. If the bubble is allowed to grow unimpeded to its equilibrium size, then the collapse may result in a second acoustic pulse as well as the emission of a water jet. This water jet results from the walls of the dense liquid colliding when the bubble collapses completely.

[0059] In some instances, calcium lesions may require high pressures (sometimes as high as 10-15 or even 20 atm) to break up the calcified plaque and push it back into the vessel wall. The system includes a pulse generator that is coupled to the proximal ends of insulated wires that provide one or more voltages to the shock wave generator. As a voltage is applied across an electrode pair by the pulse generator, each pulse initially ionizes the conductive fluid inside the balloon **26** to generate small gas bubbles around the shock wave generators that insulate the electrodes. Subsequently, a plasma arc forms across a gap between the electrodes of the electrode pairs, generating a low impedance path where current flows freely. The heat from the plasma arc heats the conductive fluid to generate a rapidly expanding vapor bubble. The expansion and collapse of the vapor bubble generates a shock wave (cavitation wave) that radiates outwardly through the balloon **26** and then through the blood to the calcified lesion proximate to the balloon **26**.

[0060] In some embodiments, the electrodes in the catheter may be activated using at least one packet (also referred to as a burst pulse, operating in burst mode operation) comprising a plurality of sub-pulses. FIGS. **6**A and **6**B illustrate exemplary graphs of a plurality of sub-pulses (top) and non-burst pulses (bottom), according to some embodiments. The duration of a sub-pulse **702** may be less than the duration of a non-burst pulse **52**, as shown in the figure. The sub-pulses **702** within a packet **704** may be delivered in rapid succession (e.g., with a frequency of 100 Hz-10 kHz). The packets **704** may be delivered in bursts. The method of generating a shock wave in a shock wave catheter system comprises applying a plurality of sub-pulses **702** within a packet **704**. In some aspects, a sub-pulse **702** generates a shock wave. The time between adjacent sub-pulses **702** (non-active sub-pulse period (time between the end of a sub-pulse and the start of an adjacent sub-pulse)) may be less than the time between adjacent packets **704** (non-active packet period (time between the end of a packet and the start of an adjacent packet)).

[0061] The frequency of the sub-pulses **702** within a packet **704** and/or the frequencies of the packets **704**, may be determined based on the properties of the calcium and/or tissue being affected by the catheter system. For example, a higher (increased) frequency (of the sub-pulses **702** or of the packets **704**) may be used when treating soft materials. Additionally or alternatively, other properties, such as number of sub-pulses **702** within a packet **704**, duration of the sub-pulses **702**, peak power of a sub-pulse **702**, electrical pulse amplitude, and/or sonic output may be adjusted when treating soft materials. In some examples, embodiments of the disclosure include varying the frequency of the sub-pulses **702** within a packet **704**. Additionally or alternatively, the frequencies of the packets **704** may vary. In some embodiments, the properties of the sub-pulses **702** (e.g., number of sub-pulses **702** within a packet **704**, the duration of at least one sub-pulse **702**, the peak power of at least one sub-pulse **702**, frequency of the sub-pulses **702**, electrical pulse amplitude, sonic output, etc.) may be determined based on the properties of the calcium and/or tissue affected (treated) by the catheter system. For example, the properties of the sub-pulses and/or packets may be determined based on the hardness, thickness, acoustic properties (e.g., acoustic impedance), etc. of the calcium/tissue. A controller (e.g., controller **990** of FIG. **9**) may be configured to control the properties of the sub-pulses **702**.

[0062] FIG. **7** illustrates an exemplary graph comparing non-burst (non-packet) pulses **52** (top) and a plurality of sub-pulses **702** (bottom), according to some embodiments. The non-burst pulse **52** (top) may comprise a single pulse, having a 1 μs pulse width, that is activated at a repetition rate of 250-1000 mS. The non-burst pulses may have amplitudes as low as 500 volts, or in the range of 1000 volts to 10,000 volts. The energy pulses may be delivered at a slow frequency, e.g., 1-4 Hz.

[0063] A packet **704** (bottom) may comprise a plurality of sub-pulses **702**. In some embodiments, a sub-pulse **702** may have a duration (pulse width) of 1 μs or shorter. The packet **704** may have a 20 μs (or longer) duration and a 50% duty cycle, for example. In some embodiments, a packet **704** comprises 10 sub-pulses **702**. In some embodiments, a packet **704** may be activated at a repetition rate of 250-1000 mS (non-active packet period). A first sub-pulse **702**A may break up a first amount of calcium, and additional sub-pulses **702**B, **702**C, etc. may break up the calcium even more.

[0064] A packet **704** may comprise a plurality of sub-pulses **702** delivered in rapid succession. For example, a packet **704** may comprise 24 sub-pulses **702**. In some embodiments, the packets **704** may have a frequency similar to pulse **52** (e.g., frequency of 1-4 Hz). In some embodiments, the frequency of the sub-pulses **702** may be at least 100 times more than the frequency of the pulse **52** or packet **704**.

[0065] The peak power of a sub-pulse **702** may be higher but its energy may be less compared to the peak power and energy of a non-burst pulse **52**. As a result, less energy is required for a sub-pulse **702** than a non-burst pulse **52** for a given amount of average power. FIG. **8**A illustrates an example comparison of power and energy (area under the power vs. time curve) for a packet **704** comprising sub-pulses **702** and a non-burst pulse **52**A. As shown in the figure, the sub-pulses **702** of the packet **704** may comprise the same peak power as the non-burst pulse **52**A. For a given amount of time (e.g., 0.5 seconds), the energy delivered for the packet **704** may be less (e.g., half) than the energy delivered for the non-burst pulse **52**A. For the same amount of energy delivered, a greater amount of time may be required to deliver a packet **704** of sub-pulses **702** than to deliver a non-burst pulse **52**A.

[0066] FIG. **8**B illustrates an example comparison of power and energy for a packet **704** comprising sub-pulses **702** and a non-burst pulse **52**B. As shown in the figure, the sub-pulses of the packet **704** may comprise a higher peak power compared to the peak power **811** of the non-burst pulse **52**B, where the same energy may be delivered in the same amount of time.

[0067] Referring back to FIG. **7**, sub-pulse **702**A may have a peak power **706**A and duration **708**A. The duration **708**A may be shorter than the duration **718** of a non-burst pulse **52**. In some embodiments, the average power of the sub-pulses **702** in a packet **704** may be substantially the same as the average power of a non-burst pulse **52**. The time between adjacent sub-pulses (e.g., sub-pulse **702**A and **702**B) may be referred to as the non-active sub-pulse period **705**A. During the non-active sub-pulse period, the power may be reduced to a non-active sub-pulse power level **726**. One or more non-active sub-pulse voltages may be applied. In some embodiments, the system may not apply one or more voltages during the non-active sub-pulse period. The non-active sub-pulse power level **726** may be less than the peak power **706**A; for example, the non-active sub-pulse power level **726** may be 30%, 40%, 50%, etc. (including a non-active power level of 0 Watts) of the peak power **706**A. The peak power **706**A may be higher than the peak power **716** of a non-burst pulse **52**. In some embodiments, the peak power **706**A of a sub-pulse **702**A may be 250 kW or higher.

[0068] The higher peak power **706** of the sub-pulse **702** may result in an electrical arc forming across the electrodes **22** and **24** faster, which may result in a bubble being generated faster. The bursts of sub-pulses **702** with shorter pulse duration **708** may have the same energy, but delivered in a shorter amount of time (e.g., within the duration of a packet **704**), leading to faster and more effective therapy. In some instances, the higher frequency (e.g., 100 Hz-10 kHz) of the sub-pulses **702** may also lead to more effective therapy, particularly when treating soft materials, such as soft tissue.

[0069] Examples of the disclosure include a shock wave catheter system configured to operate in burst mode operation and non-burst mode operation. The catheter system can be capable of switching between burst mode operation and non-burst mode operation. In some aspects, in burst mode operation, a plurality of sub-pulses, e.g., having a frequency of 100 Hz-10 kHz, duration of 1 μs or shorter, peak power of at least 250 kW or a combination thereof, within a packet are delivered in rapid succession. In some aspects, in non-burst mode operation, packets, e.g., having a frequency of 1-4 Hz, duration longer than 20 μs, peak power of less than 250 kW, or a combination thereof, are delivered. The catheter system may include a user control input that allows a user to select the mode of operation. For example, the user may select the burst mode operation when treating a first material, and then switch to the non-burst mode operation when treating a second material. The first material may be different from the second material, in some examples, such as the first material being softer tissue than the second material. In this manner, the catheter system may be able to deliver more effective therapy by selectively targeting different types of tissues using different operations.

[0070] Embodiments of the disclosure include burst mode operation using a multiple stage generator circuit. FIG. **9** illustrates a schematic diagram of a circuit for burst mode operation of an IVL catheter, according to some embodiments. The circuit comprises a sub-pulse capacitor C**2**, a reservoir capacitor C**1**, and a source **40**. Capacitor C**1** may be a reservoir capacitor that stores energy for a packet **704** (burst energy delivery), and capacitor C**2** may be a sub-pulse capacitor that stores and delivers energy for one or more sub-pulses **702**. In some aspects, the capacitance of the reservoir capacitor C**1** may be greater than the capacitance of the sub-pulse capacitor C**2** (e.g., the capacitance of C**1** may be more than twice the capacitance of C**2**). The source **40** can charge the capacitor C**1**, electrically coupled to the source **40**. The sub-pulse capacitor C**2** may be configured to store and deliver energy for one or more (e.g., each) sub-pulses **702**. An electrode of the emitter **925** is coupled to the capacitor C**1**, and an electrode of the emitter **925** is coupled to the transistor **972**. The source **942** provides a low voltage to the emitter **925** when coupled via transistor **972**. The energy of the sub-pulses **702** may be delivered across an emitter **925** (the region of an electrode assembly where the current transmits across the electrode pair **25**) for generating a shock wave. The circuit includes a transistor **972** that may couple an electrode **24** of the emitter **925** to a sense resistor **974** based on the pulse signal **970**. The resistance of the resistor **974** may limit the charge current for charging the capacitor C**1**. Coupling the electrode **24** to the sense resistor **974** (e.g., when the pulse signal **970** is high) may lead to a voltage drop across the electrodes **22** and **24** for pulse generation. Applying a high voltage to the electrode pair **25** may cause an electrical current to transmit across the spark gap of the catheter **20**.

[0071] In some embodiments, operation of the circuit may comprise a plurality of operation modes, including sub-pulse mode and charge mode. In some embodiments, the catheter may operate in sub-pulse mode and charge mode at different times. In sub-pulse mode, energy from the sub-pulse capacitor C**2** is transferred to the emitter **925**. In charge mode, the reservoir capacitor C**1** charges the sub-pulse capacitor C**2**. The multi-stage generator circuit comprises a switch **986** that is open during sub-pulse mode, electrically decoupling the reservoir capacitor C**1** from the emitter **925** so that the source **40** can charge the reservoir capacitor C**1**.

[0072] The switch **986** may be opened or closed depending on the operation mode. During sub-pulse mode, the switch **986** may be open, and the emitter **925** may be electrically coupled to the sub-pulse capacitor C**2** and the source **942**. During charge mode, the switch **986** may be closed, electrically coupling the source **40** to the sub-pulse capacitor C**2**. The transistor **972** may be open during charge mode, thereby allowing source **40** to charge the sub-pulse capacitor C**2**.

[0073] The inclusion of a reservoir capacitor C**1** and a sub-pulse capacitor C**2** helps reduce the amount of decrease in peak power for subsequent sub-pulses **702** within a packet. The reservoir capacitor C**1** is configured to provide energy to the sub-pulse capacitor C**2** during charge mode to help maximize the amount of energy stored on the sub-pulse capacitor C**2** before the beginning of a sub-pulse **702**, thereby allowing the energy delivered to the emitter **925** (during sub-pulse mode) to be close to or the same as the maximum energy storage capabilities of the sub-pulse capacitor C**2** for more than one sub-pulse **702**.
[0074] The sense resistor **974** may be a current sense resistor coupled to a sense amplifier **976**. The sense amplifier **976** may generate a sense signal **968** when the current flowing through the electrodes **22** and **24** of the emitter **925** reaches a predetermined current limit (e.g., 10-100 amps). The sense signal **968** may be a current signal sent to the controller **990** when a pulse has been detected. In some embodiments, the sense resistor **974** may control the flow of current, limiting the maximum peak output current. For example, the sense resistor **974** may be a ballast resistor that changes its resistance based on the flow of current.
[0075] The reservoir capacitor C**1** may be configured to store energy, e.g., for burst energy delivery, when the demands require more energy than stored in sub-pulse capacitor C**2** and/or delivery of energy from the source **40** alone is not fast enough for energy delivery during a given sub-pulse **702**. The source **40** may be coupled to a reservoir capacitor C**1** for charging the reservoir capacitor C**1**. The reservoir capacitor C**1** may also be coupled to ground. A switch **986** selectively couples the reservoir capacitor C**1** to the sub-pulse capacitor C**2** to transfer the energy stored in the reservoir capacitor C**1** to the sub-pulse capacitor C**2** and/or the emitter **925**. The controller **990** controls the switch **986** using a charge signal **988** such that the switch **986** is closed when charging the sub-pulse capacitor C**2** (charge mode). In some embodiments, the switch **986** is closed prior to each sub-pulse **702**.
[0076] A controller **990** (e.g., microprocessor, a microcontroller, a field programmable gate array (FPGA), etc.), or other similar control circuitry (such as a gate array), controls the overall operation of the catheter system. The controller **990** may receive a button signal coupled to a switch **992**. In some embodiments, the button signal may be generated based on a user control input (e.g., a momentary button press) to control the delivery of the sub-pulses **702** and/or packets **704**. For example, a high voltage may be delivered to electrode pair **25** when the button signal is high (has a value greater than or equal to a threshold value), closing the switch. The button signal may be high due to, e.g., a user (physician, operator, etc.) pressing a button on the catheter system. In some embodiments, the high voltage may be delivered for a pre-determined duration (e.g., the duration of a packet **704**, the duration **708** of a sub-pulse **702**, etc.). In some embodiments, the high voltage may be delivered for a duration based on the duration of the button press. The controller **990** may stop delivery of the high voltage when the HV control signal **993** is low, for example. The HV control signal **993** may be low (has a value less than a threshold value) when the pre-determined duration has elapsed, when the physician or operator is no longer pressing the button to the catheter system.
[0077] The circuit **900** comprises a source **942** that may be a low voltage power supply (LVPS). The source **942** may be used to bias the emitter **925** to generate one or more gas bubbles, e.g., in a fluid surrounding the electrode **22**, electrode **24**, or both. In some embodiments, the source **942** may keep the current flowing through the emitter **925** low between adjacent sub-pulses **702** (the non-active sub-pulse period **705**A within a packet **704**, when a non-active sub-pulse power level **726** is being applied to the electrode pair **25**). The source **942** may be coupled to a diode **944** that prevents the pulse current flowing through the source **942** and a resistor **946** that limits the current from the source **942**. The source **942** may provide a low voltage based on a LV control signal **994** from the controller **990**.
[0078] FIG. **10** illustrates an exemplary flow chart for burst mode operation of an IVL, according to some embodiments. At step **1002**, the catheter system is in an initial state. In some embodiments,

in the initial state, a physician or operator is not pressing a button, as indicated by the button signal to the controller **990**. The switch **986** is open, preventing the reservoir capacitor C**1** from discharging stored charge (transferring energy to the sub-pulse capacitor C**2** and/or emitter **925**). The transistor **972**, operating in accordance with the pulse signal **970**, does not electrically couple the emitter **925** to the sense resistor **974**. The HV control signal **993** from the controller **990** to the source **40** indicates that the source **40** should not be providing power (e.g., the HV control signal **993** is low), and the LV control signal **994** from the controller **990** to the source **942** indicates that the source **942** should not be providing power (e.g., the LV control signal **994** is low). No energy is provided to the electrode pair of the emitter **925**.

[0079] At step **1004**, the catheter system may receive a button signal indicative of a button press from a physician or an operator. In some embodiments, the button signal may be high when the physician or operator is pressing a button on the catheter system and low when the physician or operator is not pressing a button. At step **1006**, the source **942** may apply a low voltage to the emitter **925**, and at step **1008**, the catheter system may wait a time period for a bubble to generate. In some embodiments, the source **942** may apply a low voltage to the electrode **22** and electrode **24** of the emitter **925** in response to the controller **990** sending a corresponding LV control signal **994** (e.g., the LV control signal **994** is high).

[0080] At step **1010**, the controller **990** may send an HV control signal **993** that causes the source **40** to output a high power. For example, the HV control signal **993** may be high. The source **40** may provide a high power to charge the reservoir capacitor C**1**.

[0081] The controller **990** may then close the switch **986** by using the charge signal **988**. Closing the switch **986** after the reservoir capacitor C**1** has been charged may cause the sub-pulse capacitor C**2** to charge (step **1012**). The controller **990** opens the switch **986** to stop charging the sub-pulse capacitor C**2**.

[0082] At step **1014**, the energy stored in the sub-pulse capacitor C**2** is applied to the emitter **925**. This stored energy may be applied via the pulse signal **970** from the controller **990** causing the transistor **972** to electrically couple the electrode **24** of emitter **925** to the sense resistor **974**. The power is applied to the emitter **925** for the duration of a sub-pulse **702**.

[0083] When there is a certain amount of current from the emitter **925** through the transistor **972** through the sense amplifier **976**, a sense signal **968** is output to the controller **990**. The controller **990** outputs the pulse signal **970** that causes the transistor **972** to electrically decouple the emitter **925** and the sense resistor **974**. This electrical decoupling between the emitter **925** and the sense resistor **974** stops energy from being applied to the emitter **925**, returning to a non-active sub-pulse power level (step **1016**).

[0084] At step **1018**, a determination is made whether all of the sub-pulses **702** of the packet **704** have been executed. If yes, then the catheter system waits for a non-active packet period (step **1020**). If no, then the catheter system repeats steps **1006** to **1016** for the next sub-pulse **702**.

[0085] FIG. **11** illustrates a graph of an exemplary burst mode operation of a multi-stage generator circuit for a packet, according to some embodiments. The multi-stage generator circuit may be more effective at breaking up calcium than the single-stage generator circuit. Although FIG. **11** illustrates a packet **704** as comprising 10 sub-pulses **702**, embodiments of the disclosure may include any number of sub-pulses **702** within a packet **704**.

[0086] The sub-pulses **702** in a packet **704** may have substantially the same peak power. For example, the first sub-pulse **702**A may have a peak power of about 340 kW, the second sub-pulse **702**B may have a peak power of about 340 kW, the third sub-pulse **702**C may have a peak power of about 340 kW, etc.

[0087] FIG. **12** illustrates a graph of exemplary charge and sub-pulse modes, and FIG. **13** illustrates an exemplary sub-pulse, according to some embodiments. The multi-stage generator circuit may be operated in a plurality of operation modes, such as charge mode **1204** and sub-pulse mode **1202**. During charge mode **1204**, transistor **972** (of FIG. **9**) may be open, and the source **942** (of FIG. **9**) may apply a low voltage to the emitter **925** (of FIG. **9**). The switch **986** (of FIG. **9**) being open, allows the source **40** (of FIG. **9**) to charge the reservoir capacitor C**1**.

[0088] During sub-pulse mode **1202**, the switch **986** (of FIG. **9**) closes momentarily to charge the sub-pulse capacitor C**2**. After the sub-pulse capacitor C**2** charges, the switch **986** is open, allowing the sub-pulse capacitor C**2** to provide energy to the emitter **925** (of FIG. **9**).

[0089] As shown in the single sub-pulse of FIG. **13**, the circuit may operate in charge mode **1204** and sub-pulse mode **1202** at different times. In some embodiments, the duration of the charge mode **1204** may be greater than the duration of the sub-pulse mode **1202**.

[0090] Embodiments of the disclosure include burst mode operation using a single-stage generator circuit. FIG. **14** illustrates a simplified schematic diagram simulating an example single-stage generator circuit, according to some embodiments. A multi-stage generator circuit may comprise additional transistors and capacitors compared to the single-stage generator circuit. For example, the single-stage generator circuit may not comprise switch **986** or sub-pulse capacitor C**2**.

[0091] FIG. **15** illustrates a graph of an exemplary burst mode operation of a single-stage generator circuit for a packet, and FIG. **16** illustrates an exemplary sub-pulse, according to some embodiments. Although FIG. **15** illustrates a packet **704** as comprising 10 sub-pulses **702**, embodiments of the disclosure may include any number of sub-pulses **702** within a packet **704**, including greater than or equal to 10 sub-pulses.

[0092] Within a packet **704**, the first sub-pulse **702**A may have the highest peak power, and then the peak power(s) of subsequent sub-pulse(s) decrease. The amount of energy stored in the capacitor C**1** (of FIG. **14**) is reduced for each sub-pulse due to energy being transferred to emitter **925**. For example, the first sub-pulse **702**A may have a peak power of about 340 kW (also shown in FIG. **15**), the second sub-pulse **702**B may have a peak power of about 220 kW (the peak power of the second sub-pulse **702**B may be less than the peak power of the first sub-pulse **702**A), the third sub-pulse **702**C may have a peak power of about 150 kW (the peak power of the third sub-pulse **702**C may be less than the peak power of the second sub-pulse **702**B), etc. In some embodiments, the peak power may exhibit an exponential decay over time and/or for subsequent sub-pulses **702** within a packet **704**, as shown in FIG. **15**.

[0093] Operation of the single-stage generator circuit may be similar to the operation shown in FIG. **10**. In some embodiments, the operation of the single-stage generator may not include steps related to charging capacitors, such as charging the reservoir capacitor of step **1010** and charging the sub-pulse capacitor of step **1012**.

[0094] In some embodiments, the multi-stage generator circuit may be operated as a single stage or two stages. For example, the capacitor C**1** (of FIG. **14**) may be initially charged at the beginning of a packet **704**. Once the capacitor C**1** is charged, the transistor Q**1** may be configured to be open during the remainder of the packet **704**.

[0095] Although shock wave devices described herein generate shock waves based on high voltage applied to electrodes, it should be understood that a shock wave device additionally or alternatively may comprise a laser and optical fibers as a shock wave emitter system whereby the laser source delivers energy through an optical fiber and into a fluid to form shock waves and/or cavitation bubbles.

[0096] The electrode assemblies and catheter devices described herein may be used for treating coronary occlusions, such as lesions in vasculature, and a variety of other occlusions, such as occlusions in the peripheral vasculature (e.g., above-the-knee, below-the-knee, iliac, carotid, etc.). For further examples, similar designs may be used for treating soft tissues, such as cancer and tumors (i.e., non-thermal ablation methods), blood clots, fibroids, cysts, organs, scar and fibrotic tissue removal, or other tissue destruction and removal. Electrode assembly and catheter designs could also be used for neurostimulation treatments, targeted drug delivery, treatments of tumors in body lumens (e.g., tumors in blood vessels, the esophagus, intestines, stomach, or vagina), wound treatment, non-surgical removal and destruction of tissue, or used in place of thermal treatments or cauterization for venous insufficiency and fallopian ligation (i.e., for permanent female contraception).

[0097] In one or more examples, the electrode assemblies and catheters described herein could also be used for tissue engineering methods, for instance, for mechanical tissue decellularization to create a bioactive scaffold in which new cells (e.g., exogenous or endogenous cells) can replace the old cells; introducing porosity to a site to improve cellular retention, cellular infiltration/migration, and diffusion of nutrients and signaling molecules to promote angiogenesis, cellular proliferation, and tissue regeneration similar to cell replacement therapy. Such tissue engineering methods may be useful for treating ischemic heart disease, fibrotic liver, fibrotic bowel, and traumatic spinal cord injury (SCI). For instance, for the treatment of spinal cord injury, the devices and assemblies described herein could facilitate the removal of scarred spinal cord tissue, which acts like a barrier for neuronal reconnection, before the injection of an anti-inflammatory hydrogel loaded with lentivirus to genetically engineer the spinal cord neurons to regenerate.

[0098] FIG. **17** illustrates an exemplary of a computing system **1700**, in accordance with some examples of the disclosure. System **1700** can be a client or a server. As shown in FIG. **17**, system **1700** can be any suitable type of processor-based system, such as a personal computer, workstation, server, handheld computing device (portable electronic device) such as a phone or tablet, or dedicated device. The system **1700** can include, for example, one or more of input device **1720**, output device **1730**, one or more processors **1710**, storage **1740**, and communication device **1760**. Input device **1720** and output device **1730** can generally correspond to those described above and can either be connectable or integrated with the computer.

[0099] Input device **1720** can be any suitable device that provides input, such as a push-button switch, a touch screen, keyboard or keypad, mouse, gesture recognition component of a virtual/augmented reality system, or voice-recognition device. Output device **1730** can be or include any suitable device that provides output, such as a display, touch screen, haptics device, virtual/augmented reality display, or speaker.

[0100] Storage **1740** can be any suitable device that provides storage, such as an electrical, magnetic, or optical memory including a RAM, cache, hard drive, removable storage disk, or other non-transitory computer readable medium. Communication device **1760** can include any suitable device capable of transmitting and receiving signals over a network, such as a network interface chip or device. The components of the computing system **1700** can be connected in any suitable manner, such as via a physical bus or wirelessly.

[0101] Processor(s) **1710** can be any suitable processor or combination of processors, including any of, or any combination of, a central processing unit (CPU), graphics processing unit (GPU), field programmable gate array (FPGA), programmable system on chip (PSOC), and application-specific integrated circuit (ASIC). Software **1750**, which can be stored in storage **1740** and executed by one or more processors **1710**, can include, for example, the programming that embodies the functionality or portions of the functionality of the present disclosure (e.g., as embodied in the devices as described above)

[0102] Software **1750** can also be stored and/or transported within any non-transitory computer-readable storage medium for use by or in connection with an instruction execution system, apparatus, or device, such as those described above, that can fetch instructions associated with the software from the instruction execution system, apparatus, or device and execute the instructions. In the context of this disclosure, a computer-readable storage medium can be any medium, such as storage **1740**, that can contain or store programming for use by or in connection with an instruction execution system, apparatus, or device.

[0103] Software **1750** can also be propagated within any transport medium for use by or in connection with an instruction execution system, apparatus, or device, such as those described above, that can fetch instructions associated with the software from the instruction execution system, apparatus, or device and execute the instructions. In the context of this disclosure, a

transport medium can be any medium that can communicate, propagate or transport programming for use by or in connection with an instruction execution system, apparatus, or device. The transport computer readable medium can include, but is not limited to, an electronic, magnetic, optical, electromagnetic, or infrared wired or wireless propagation medium.
[0104] System **1700** may be connected to a network, which can be any suitable type of interconnected communication system. The network can implement any suitable communications protocol and can be secured by any suitable security protocol. The network can comprise network links of any suitable arrangement that can implement the transmission and reception of network signals, such as wireless network connections, T**1** or T**3** lines, cable networks, DSL, or telephone lines.
[0105] System **1700** can implement any operating system suitable for operating on the network. Software **1750** can be written in any suitable programming language, such as C, C++, Java, or Python. In various embodiments, application software embodying the functionality of the present disclosure can be deployed in different configurations, such as in a client/server arrangement or through a Web browser as a Web-based application or Web service, for example.
[0106] The elements and features of the exemplary electrode assemblies and catheters discussed above may be rearranged, recombined, and modified, without departing from the present invention. Furthermore, numerical designators such as “first,” “second,” “third,” “fourth,” etc. are merely descriptive and do not indicate a relative order, location, or identity of elements or features described by the designators. For instance, a “first” shock wave may be immediately succeeded by a “third” shock wave, which is then succeeded by a “second” shock wave. As another example, a “third” emitter may be used to generate a “first” shock wave and vice versa. Accordingly, numerical designators of various elements and features are not intended to limit the disclosure and may be modified and interchanged without departing from the subject invention.
[0107] As provided herein, it should be appreciated that any disclosure of a numerical range describing dimensions or measurements such as thicknesses, length, weight, time, frequency, temperature, voltage, current, angle, etc. is inclusive of any numerical increment or gradient within the ranges set forth relative to the given dimension or measurement.
[0108] It should be noted that the elements and features of the example catheters illustrated throughout this specification and drawings may be rearranged, recombined, and modified without departing from the present invention. For instance, while this specification and drawings describe and illustrate catheters having several example balloon designs, the present disclosure is intended to include catheters having a variety of balloon configurations. The number, placement, and spacing of the electrode pairs of the shock wave generators can be modified without departing from the subject invention. Further, the number, placement, and spacing of balloons of catheters can be modified without departing from the subject invention.
[0109] It should be understood that the foregoing is only illustrative of the principles of the invention, and that various modifications, alterations and combinations can be made by those skilled in the art without departing from the scope and spirit of the invention. Any of the variations of the various catheters disclosed herein can include features described by any other catheters or combination of catheters herein. Furthermore, any of the methods can be used with any of the catheters disclosed. Accordingly, it is not intended that the invention be limited, except as by the appended claims.

## Claims

**1-34**. (canceled)
**35**. A method performed by a catheter system, comprising: applying, by an energy source, a first packet of energy to an electrode pair configured to produce cavitation waves within a catheter, the first packet of energy including consecutive sub-pulses separated by a first duration; and applying,

by the energy source, a second packet of energy to the electrode pair so that the first packet of energy and the second packet of energy are separated by a second duration different than the first duration, the second packet of energy including consecutive sub-pulses separated by the first duration.
**36**. The method of claim 35, wherein the cavitation waves are produced by the electrode pair in response to the consecutive sub-pulses in each of the first packet of energy and the second packet of energy.
**37**. The method of claim 35, wherein the second duration is longer than the first duration.
**38**. The method of claim 35, further comprising: applying, by the energy source, a voltage to the electrode pair, the voltage producing a cavitation bubble in a conductive medium surrounding the electrode pair.
**39**. The method of claim 38, wherein the conductive medium comprises a liquid.
**40**. The method of claim 35, wherein the consecutive sub-pulses have the same voltage in at least one of the first packet of energy or the second packet of energy.
**41**. The method of claim 35, wherein the consecutive sub-pulses have different voltages in at least one of the first packet of energy or the second packet of energy.
**42**. The method of claim 35, further comprising: applying, by the energy source, a series of pulses to the electrode pair so that consecutive pulses in the series are separated by a third duration different than the first duration.
**43**. The method of claim 42, wherein the third duration is longer than the first duration.
**44**. The method of claim 42, wherein each of the pulses in the series has a longer pulse duration than any of the consecutive sub-pulses in the first packet of energy or the second packet of energy.
**45**. The method of claim 42, wherein each of the pulses in the series has a lower peak power than any of the consecutive sub-pulses in the first packet of energy or the second packet of energy.
**46**. The method of claim 42, wherein the catheter system is operable in at least a first mode and a second mode such that the first and second packets of energy are applied to the electrode pair while operating in the first mode and the series of pulses are applied to the electrode pair while operating in the second mode.
**47**. The method of claim 46, further comprising: receiving user input indicating a selection of the first mode or the second mode; and switching between the first mode and the second mode responsive to the user input.
**48**. A catheter system comprising: a catheter including an electrode pair configured to produce cavitation waves; an energy source; and a controller configured to: apply a first packet of energy to the electrode pair based on the energy source, the first packet of energy including consecutive sub-pulses separated by a first duration; and apply a second packet of energy to the electrode pair based on the energy source so that the first packet of energy and the second packet of energy are separated by a second duration different than the first duration, the second packet of energy including consecutive sub-pulses separated by the first duration.
**49**. The catheter system of claim 48, wherein the energy source comprises a voltage generator, a current generator, or a pulse generator.
**50**. The catheter system of claim 48, further comprising: a first capacitor configured to store and deliver energy from the energy source for each of the consecutive sub-pulses of the first and second packets of energy; and a second capacitor configured to transfer energy from the energy source to the first capacitor within the first duration.
**51**. The catheter system of claim 48, wherein the controller is further configured to: apply a voltage to the electrode pair based on the energy source, the voltage producing a cavitation bubble in a conductive medium surrounding the electrode pair.
**52**. The catheter system of claim 48, wherein the controller is further configured to: apply a series of pulses to the electrode pair based on the energy source so that consecutive pulses in the series are separated by a third duration different than the first duration.

**53**. The catheter system of claim 52, wherein the catheter system is operable in at least a first mode and a second mode such that the first and second packets of energy are applied to the electrode pair while operating in the first mode and the series of pulses are applied to the electrode pair while operating in the second mode.

**54**. A controller for a catheter system comprising: one or more processors; and a memory storing instructions that, when executed by the one or more processors, cause the controller to: apply, by an energy source, a first packet of energy to an electrode pair configured to produce cavitation waves within a catheter, the first packet of energy including consecutive sub-pulses separated by a first duration; and apply, by the energy source, a second packet of energy to the electrode pair so that the first packet of energy and the second packet of energy are separated by a second duration different than the first duration, the second packet of energy including consecutive sub-pulses separated by the first duration.